

CAPSTONE PROJECT 1

Project Proposal

1. Title of the Project

Predict Potential Spammers on Fiverr Platform using Machine Learning

2. Brief on the project

This project aims to build a binary classification model to identify potential spammer accounts on the Fiverr freelance platform. Attackers are using Fiverr to distribute malware like Agent Tesla disguised as job offers, which leads to users losing earnings and account access.

The motivation is to protect the Fiverr ecosystem by proactively flagging high-risk accounts before they can cause financial harm. This problem is important because it directly impacts platform trust and user security.

We will use the anonymized dataset from a Kaggle competition containing 51 behavioral features (X1-X51) for 458,798 users. The approach is to train a Random Forest Classifier to distinguish between 'Genuine' and 'Spammer' users based on their activity patterns.

3. Deliverables of the project

The project will deliver the following:

1. A trained Random Forest classification model capable of predicting spammer accounts.
2. A detailed model evaluation using Accuracy, ROC-AUC Score, and a Confusion Matrix, with a special focus on maximizing Spammer Recall.
3. Identification and analysis of the most important features (e.g., X19, X6, X7) that indicate spammer behavior.
4. A final project report and a Jupyter Notebook containing the complete, reproducible code.

List of questions the model is designed to answer:

- Can we predict if a new user account is a potential spammer based on its initial behavior?
- Which anonymized behavioral features are the strongest indicators of spamming activity?
- What is the trade-off between catching spammers (Recall) and falsely flagging genuine users (Precision)?

4. Resources

Data set source: Kaggle Competition - "Predict Potential Spammers on Fiverr".

URL: <https://www.kaggle.com/c/predict-potential-spammers-on-fiverr/data>

Software: Python 3, Pandas for data manipulation, Scikit-learn for machine learning, Matplotlib & Seaborn for visualization. The chosen platform is Google Colab for cloud-based execution.

References: Not compulsory for this submission as per guidelines.

5. Individual Details

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6. Milestones

| Milestones | Status |

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| Define a problem | Completed |

| Understanding the business problem | Completed |

| Get the Data | Completed |

| Explore and pre-process data | Completed |

| Choosing the python platform | Completed |

| Create Features | Completed |

| EDA | Completed |

| Create Model | Completed |

| Model Evaluation | Completed |

| Report Writing | Completed |

| Project submission | In Progress |